

Random Signal Theory–EEE 554

Lecture Notes 3

Conditional Probability Density Functions

[Reading: Kay, Sections 13.3, 13.4, 13.6]

Conditional PDF and Bayesian Filtering

- Conditional PDF of Y given X

$$f_{Y|X}(y|x) = \frac{f_{XY}(x, y)}{f_X(x)} \text{ or, for notational simplicity } f(y|x) = \frac{f(x, y)}{f(x)}$$

- Bayes' relation

Using $f(x, y) = f(y|x) f(x) = f(x|y) f(y)$

$$\Rightarrow \boxed{f(y|x) = \frac{f(x|y) f(y)}{f(x)} = \frac{f(x|y) f(y)}{\int f(x|y) f(y) \, dy}}$$

► Dynamic systems and Bayesian filtering

- Given available noisy sensor measurements, $\Phi_{1:k} = \{\Phi_1, \Phi_2, \dots, \Phi_k\}$, up to time k
e.g., measure signal $\phi_c(t)$ and sample $\phi[k] = \phi_c(kT_s)$; each sample is a RV $\Phi_k = \phi[k]$
- Goal: estimate unknown system parameter (or state) Z_k at time k using $\Phi_{1:k}$
- State space representation

transition model: $Z_k = g(Z_{k-1}) + U_{k-1}$

measurement model: $\Phi_k = h(Z_k) + V_k$

- $g(\cdot)$: function that relates unknown state Z_k at time k to state Z_{k-1} at time $k - 1$
- U_{k-1} : possible error RV due to model mismatch
- $h(\cdot)$: function that relates signal Φ_k at time k to unknown state Z_k at time k
- V_k : measurement noise sample RV

Example: estimate position and velocity of vehicle moving at constant velocity

– Unknown state Z_k :

Cartesian coordinates of vehicle position (x_k, y_k) and velocity $(\dot{x}_k, \dot{y}_k)$ at time step k

$$Z_k = \begin{bmatrix} x_k \\ y_k \\ \dot{x}_k \\ \dot{y}_k \end{bmatrix} = \begin{bmatrix} 1 & 0 & \delta t & 0 \\ 0 & 1 & 0 & \delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_{k-1} \\ y_{k-1} \\ \dot{x}_{k-1} \\ \dot{y}_{k-1} \end{bmatrix} + \begin{bmatrix} u_{x_{k-1}} \\ u_{y_{k-1}} \\ u_{\dot{x}_{k-1}} \\ u_{\dot{y}_{k-1}} \end{bmatrix}$$

e.g., position at time k obtained as $x_k = x_{k-1} + \delta t \dot{x}_{k-1} + u_{x_{k-1}}$

where δt is time between steps k and $k - 1$

and $u_{x_{k-1}}$ models error due to possible acceleration

– Radar measurement Φ_k : distance (range) r_k and range rate $\dot{r}_k$

$$\Phi_k = \begin{bmatrix} r_k \\ \dot{r}_k \end{bmatrix} = \begin{bmatrix} \sqrt{x_k^2 + y_k^2} \\ \frac{\dot{x}_k x_k + \dot{y}_k y_k}{r_k} \end{bmatrix} + V_k$$

To estimate state Z_k using measurements $\Phi_{1:k}$, must estimate conditional PDF

PDFs involved in state estimation

- prior PDF: $f(z_0 | \phi_0) = f(z_0)$
- conditional PDF for transition: $f(z_k | z_{k-1})$
- a-priori PDF: $f(z_k | \phi_{1:k-1})$
- conditional PDF for measurement or likelihood PDF: $f(\phi_k | z_k)$
- posterior PDF: $f(z_k | \phi_{1:k})$
- filtered estimate obtained as the mean of posterior PDF (conditional mean)

Estimate relies on:

- Markovian process of state transition:
probability of future state depends only on present state; first order Markov process
satisfies memoryless property: $f(z_k | z_{k-1}, \dots, z_1, z_0) = f(z_k | z_{k-1})$
- Current measurement depends only on current state
conditionally independent of all other states

$$f(\phi_k | z_k, z_{k-1}, \dots, z_0) = f(\phi_k | z_k)$$

Bayesian filtering steps $\Rightarrow$ recursively estimate state at time k given measurements

- Assume prior $f(z_0 | \phi_0) = f(z_0)$
- Assume $f(z_{k-1} | \phi_{1:k-1})$ is available at time $k-1$
- Prediction step: obtain $f(z_k | \phi_{1:k-1})$ using Chapman-Kolmogorov equation

$$\begin{aligned} f(z_k | \phi_{1:k-1}) &= \int f(z_k | z_{k-1}, \phi_{1:k-1}) f(z_{k-1} | \phi_{1:k-1}) \mathrm{d}z_{k-1} \\ &= \int f(z_k | z_{k-1}) f(z_{k-1} | \phi_{1:k-1}) \mathrm{d}z_{k-1} \end{aligned}$$

where $f(z_k | z_{k-1}, \phi_{1:k-1}) = f(z_k | z_{k-1})$ (from Markov property)

- Update step: use measurements to obtain likelihood $f(\phi_k | z_k)$

Recall general rule:
$$f(y|x) = \frac{f(x|y) f(y)}{f(x)} = \frac{f(x|y) f(y)}{\int f(x|y) f(y) \, dy}$$

- Posterior PDF using Baye's

$$f(z_k | \phi_{1:k}) = \frac{f(\phi_k | z_k) f(z_k | \phi_{1:k-1})}{f(\phi_{1:k})} = \frac{f(\phi_k | z_k) f(z_k | \phi_{1:k-1})}{\int f(\phi_k | z_k) f(z_k | \phi_{1:k-1}) \, dz_k}$$

Conditional mean provides state estimate from estimated posterior PDF:

$$\hat{z}_k = E[z_k | \phi_{1:k}] = \int z_k f(z_k | \phi_{1:k}) \, dz_k$$

Computation of predict and update equations

- Sequential Monte Carlo methods (simulation-based, e.g., particle filter)
if models nonlinear and/or non-Gaussian RVs
- Kalman filter (KF): linear models and Gaussian RVs $U_k \sim \mathcal{N}(0, C_u)$, $V_k \sim \mathcal{N}(0, C_v)$
transition model: $Z_k = G Z_{k-1} + U_{k-1}$ measurement model: $\Phi_k = H Z_k + V_k$
- At time $k-1$, assume have state estimate $\hat{z}_{k-1|k-1}$ and its covariance matrix $P_{k-1|k-1}$
- Prediction from time $k-1$ to k :
$$\hat{z}_{k|k-1} = G \hat{z}_{k-1|k-1}$$
$$P_{k|k-1} = C_u + G P_{k-1|k-1} G^T$$
$$S_k = H P_{k|k-1} H^T + C_v$$
$$K_k = P_{k|k-1} H^T S_k^{-1} \text{ (KF gain)}$$
- Update at time k :
$$\hat{z}_{k|k} = \hat{z}_{k|k-1} + K_k (\phi_k - H \hat{z}_{k|k-1})$$
$$P_{k|k} = P_{k|k-1} - K_k H P_{k|k-1}$$